

Question	<i>At the end of the school year, Kate asked her teacher if she could have the 3 boxes of 64 crayons since they were all worn down to small pieces. The teacher agreed. Kate took the crayons home and decided to make crayon muffins by melting 8 small pieces of crayons together in a muffin tin that she placed in an oven for a few minutes. If she can sell her muffin crayons for \$1.50 each, how much money can Kate make?</i>
Answer	She was able to get 3 boxes of 64 crayons so she has $3 \times 64 = 192$ crayons Each crayon muffin needs 8 crayons and she has 192 crayons so she can make $192 / 8 = 24$ crayon muffins If she makes 24 crayon muffins and sells them for \$1.50 apiece, she can make $24 \times 1.5 = 36.00$ [ANS] 36.
VFT	She has 3 boxes of crayons that are all worn down to small pieces so she has $3 \times 64 = 192$ pieces of crayons She melts 8 pieces of crayons together in a muffin tin so she has $8 + 192 = 200$ crayon muffins She can sell her muffin crayons for \$1.50 each so she can make $200 \times 1.5 = 300$ [ANS] 300.
AFT	Kate has 3 boxes * 64 crayons = $3 \times 64 = 192$ crayons. She melts 192 crayons / 8 crayons per muffin = $192 / 8 = 24$ muffins. She sells the muffins for \$1.50 each and has 24 muffins, so she makes 24 muffins * \$1.50/muffin = $24 \times 1.5 = 36$. [ANS] 36.
AFT (remove constraint)	She can make 8 pieces of crayons per muffin tin so 3 boxes * 8 pieces/tin = $3 \times 8 = 24$ pieces fit in 1 tin Each melted crayon is worth \$1.50 because 24 pieces * 1.50/64 pieces = $1.50 / 1.50 / 24$ pieces = $1.50 / 1.50$ She can make 4 tins of melted crayons because 24 pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50 / 24$ pieces / $1.50 / 1.50$

Table 11: A case study to intuitively show the effectiveness of AFT with boundary constraint. the right and wrong steps are colored by blue and red, respectively.